

1. Using the life table, obtain

a) e_{30}

b) e^o_{30}

c) e_{40}

d) e^o_{45}

2. Suppose that $s(x) = 1 - x/10$, $0 \leq x \leq 10$.

(a) Find e_x .

(b) Using life table terminology, find P_2 , q_3 , ${}_3P_7$, and ${}_2q_7$.

3. Complete the entries in the following table:

x	e_x	d_x	P_x	q_x
0	100,000			
1	99,499			
2	98,995			
3	98,489			
4	97,980			
5	97,468			

4. Given

$$f(t) = \frac{20-t}{200}, \quad 0 \leq t < 20$$

Find e^o_5

5. Given the following excerpt of a life table

x	e_x
50	100,000
51	99,900
52	99,700
53	99,500
54	99,100
55	98,500

(a) Calculate d_{52}

(b) Calculate ${}_2|q_{50}$

(c) Assuming uniform distribution of deaths between integer ages, calculate the value of ${}_{4.3}P_{50.4}$